

SHAKTI-CHAIN

A vehicle-to-grid marketplace with incentive-compatible clearing and Layer-2 settlement

Shakti-Chain is a working software platform from BITS Pilani that lets electric-vehicle owners, fleets, and buyers trade energy in 15-minute rounds. Price discovery uses a McAfee double auction: truthful per-unit price reporting is a dominant strategy, no matched participant loses money, and the operator does not require an external subsidy. Settlement uses an ERC-20 token on Polygon with a mint-on-delivery, burn-on-redemption lifecycle that holds a 1:1 kWh peg.

The gap this fills

India's FAME programme targets 30% EV penetration by 2030: on the order of 50 million battery-equipped vehicles and roughly 300 GWh of parked storage. Peak demand is projected above 340 GW. Bidirectional inverters and smart meters already exist at pilot scale. What is missing is a settlement and price-discovery layer that can clear high-frequency, low-value V2G trades without a trusted intermediary.

- A typical discharge event is ₹15–40. UPI merchant fees and batch latency consume a large share of that value and cannot hold conditional escrow.
- Layer-1 blockchain settlement (Brooklyn Microgrid class) has historically cost more than the energy being traded.
- Fixed DISCOM tariffs do not elicit truthful bids. Prior V2G auctions either lacked formal incentive proofs or did not settle on-chain.

What is different

Property	What Shakti-Chain does
Clearing	McAfee double auction. Price is set by excluded marginal traders, so winners cannot move their own price.
Guarantees	Dominant-strategy incentive compatibility (per-unit price), ex-post individual rationality, strong budget balance.
Token	SHAKTI ERC-20: minted only on confirmed delivery, burned on redemption. Designed 1 token = 1 kWh.
Settlement rail	Polygon Layer-2. Matching stays off-chain; only settlement and the token lifecycle go on-chain.
India calibration	Agent valuations follow city load distributions for Delhi, Mumbai, Bangalore, Chennai, and Kolkata (POSOCO).

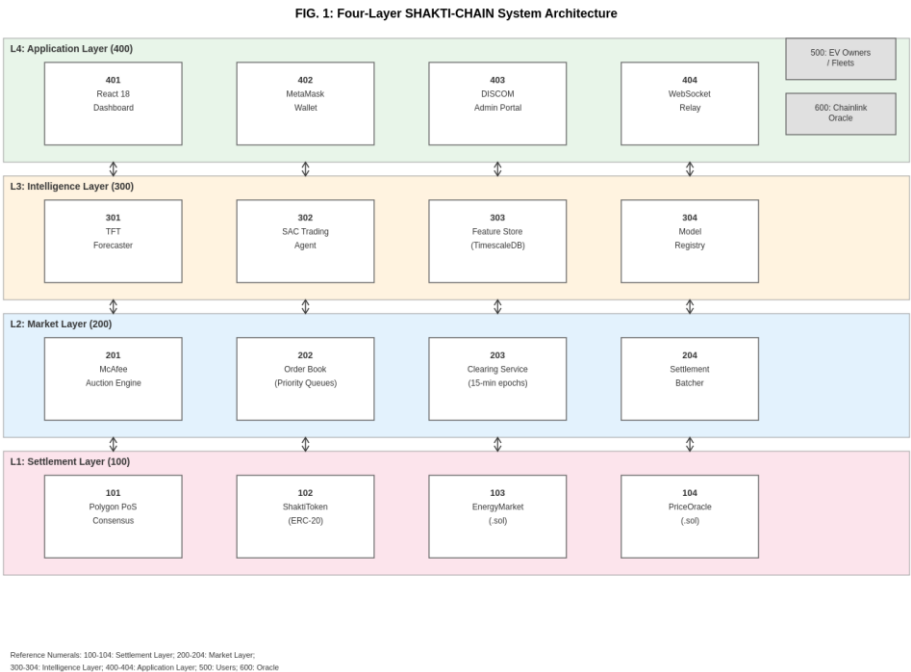
Measured results (simulation)

Pre-registered evaluation on 2,000 heterogeneous agents across five Indian metropolitan service areas. These figures are simulation measurements, not a live-grid trial.

Quantity	Result	Note
Allocative efficiency	97.3%	2.7 pp below uniform-price; uniform-price is not DSIC
Gross ROI vs time-of-use	18.7%	About 10–13% post-tax under current VDA rules
kWh peg deviation	0.007%	3,000 redemption events
Feeder hosting-capacity wrapper	0.16%	Average welfare curtailed after clearing
5-agent coalition surplus	12.3%	Failed the 10% pre-registered cap; known McAfee limit

Working demonstration

The marketplace application is running: React dashboard, FastAPI market engine, McAfee clearing, demo ledger (no wallet required). A visitor can place buy and sell orders, watch a round clear, stake tokens, and run a city-calibrated simulation. Source and runbook: github.com/amitduabits/shaktichain (see DEMO.md).



Who this is for

- DISCOMs — a clearing layer above existing metering, with a feeder hosting-capacity wrapper after each round.
- Charge-point operators and EV OEMs — a software path to monetise bidirectional chargers without building an auction from scratch.
- Fleet owners — a way to sell parked energy during evening peak under a rule that does not punish honest bidding.
- Aggregators and energy-service companies — a licenceable matching and settlement stack, not a consumer app.

Status and what we are asking

Software demo	Working. Login, order ticket, clearing, staking, city simulation.
Intellectual property	Copyright BITS Pilani. Indian patent application filed. Not open-source. Not MIT.
Paper	Under review, IEEE Transactions on Smart Grid (target).
Not yet	Live DISCOM billing integration; physical charger control; voltage and transformer thermal limits.

BITS Pilani is seeking one distribution-circle pilot, a sponsored research agreement, or a technology licence. Typical first step: a 20-minute live demo, then a written term sheet through the Institute technology-transfer process. There is no informal commercial use of this code.

Investigators

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